Sandesh S. Kalantre

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EDUCATION

Stanford University, USA *June 2022 - Sept. 2026*
PhD in Physics, Knight-Hennessy Fellow, Advisor: Prof. David Goldhaber-Gordon
Thesis: Microwave Probes of Quantum Matter

University of Maryland (UMD), College Park *Aug 2018 - May 2022*
Lanczos Graduate Fellow (*transferred to Stanford*) Advisor: Prof. James R. Williams

Indian Institute of Technology (IIT) - Bombay, India *Aug 2014- May 2018*
B.Tech in Engineering Physics with Honors and Minor in CS, Institute Silver Medal

EMPLOYMENT

Postdoc, UC Berkeley *Sep 2026 - Present*
Suleymanzade Group: Hybrid mm-wave interfaces between atoms and superconducting circuits

RESEARCH HIGHLIGHTS

- **First ML-based autotuning of a quantum dot device.** Developed CNN classifiers and a closed-loop algorithm that autonomously tuned a semiconductor quantum dot to the single-electron regime in situ, an approach now adopted across the field. Kalantre et al., *npj Quantum Information* (2019)
- **Microwave spectroscopy of a thin film quantum spin liquid candidate.** Built a NbTiN flip-chip resonator platform to measure spin resonance in epitaxial $TbInO_3$, enabling GHz probes of frustrated magnetism in films too small for conventional ESR. Kalantre et al., arXiv:2603.14545 (under review).
- **Fermiology of a candidate chiral superconductor in rhombohedral graphene.** Measured quantum oscillations in rhombohedral tetralayer graphene and showed that the superconductor's normal state is a complex "multitone" state rather than a simple quarter metal, constraining theories of the pairing. Kalantre et al., arXiv:2606.05356 (under review).
- **cQED architecture for superfluid stiffness in 2D superconductors.** Designed a microwave platform to extract kinetic inductance and superfluid stiffness of van der Waals superconductors at GHz frequencies and mK temperatures.
- **Resolved puzzling Shapiro measurements in driven graphene Josephson junctions.** Developed a non-linear oscillator framework that explained anomalous Shapiro steps in graphene Josephson junctions, reconciling microwave response with junction parameters. Kalantre et al., *Physical Review Research* (2020).

HONORS AND AWARDS

- Knight-Hennessy Fellowship ($60,000/yr) *2023-2026*
  For visionary & collaborative leaders addressing complex real-world challenges at Stanford
- Lanczos Graduate Fellowship *2018-2020*
  Awarded to top incoming 2-3 students ($36,000/yr) at Joint Quantum Institute, UMD
- Institute Silver Medal for graduating at the top of Engineering Physics program, IIT Bombay *2018*
- S. N. Bose Scholarship ($2000) awarded to top 50 students from India in STEM *2017*
- Institute Academic Award and R.P. Singh Memorial Award, IIT Bombay *2015-2017*
- ISTernship at IST Austria ($2500) awarded to around 40 students worldwide 2016
- National Initiative on Undergraduate Science (NIUS) Program in Physics *2014-2015*
- Kishore Vigyan Protsahan Yojana (KVPY), financial award for basic science high-school students *2013*
- 2 Gold and 1 Silver medals at International Olympiads on Astronomy and Astrophysics *2011-2013*

PREPRINTS AND PUBLICATIONS

Google Scholar | **600+ citations** | **h-index = 11**

17. **Sandesh S. Kalantre**, Ben H. Alexander, Julian May-Mann, Jonah Herzog-Arbeitman, Marisa Hocking, Qingrui Cao, Kenji Watanabe, Takashi Taniguchi, David Goldhaber-Gordon, Andrew J Mannix, Trithep Devakul, Yves H Kwan, Daniel E Parker, Aaron Sharpe, Fermiology and the Candidate Chiral Superconductor in Rhombohedral Tetralayer Graphene, arXiv:2606.05356 *(under review at Nature)*

16. Owen I Sheekey, Trevor B Arp, Benjamin A Foutty, Ruoxi Zhang, Tixuan Tan, Ludwig FW Holleis, Yi Guo, **Sandesh S. Kalantre**, Canxun Zhang, Mark Zakharyan, David Gong, Aidan Keough, Youngjoon Choi, Ysun Choi, Siyuan Xu, Tian Xie, Ben Hodder Alexander, Marisa Hocking, Qingrui Cao, Martin E Huber, Takashi Taniguchi, Kenji Watanabe, Chenhao Jin, Etienne Lantagne-Hurtubise, Aaron Sharpe, Trithep Devakul, Andrea F Young, Visualizing orbital magnetism in electron doped rhombohedral multilayer graphene, arXiv:2605.30316 *(under review at Nature)*

15. Haotian Su, Sauviz Alaei, Ashildur Fridriksdóttir, Sydney Fultz-Waters, Hyeonhu Bae, Xiangjin Wu, Yufei Zhao, Donglai Zhong, Tara Peña, **Sandesh S. Kalantre**, Andrew Barnum, Gregory Pitner, Sam Vaziri, Xinyu Bao, David Goldhaber-Gordon, Paul McIntyre, Shan Wang, Aaron Sharpe, Yuri Suzuki, Binghai Yan, Eric Pop, Unconventional resistivity reduction in NbP semimetal nanoribbons, *(under review at Nature)*

14. **Sandesh S. Kalantre**, Johanna Nordlander, Margaret A. Anderson, Julia A. Mundy, and David Goldhaber-Gordon, Microwave spin resonance in epitaxial thin films of spin liquid candidate $TbInO_3$ arXiv:2603.14545, 2026 *(under review - Physical Review X)*

13. Anjolaoluwa Bamtefa, **Sandesh S. Kalantre**, Margaret A. Anderson, Austin J. Akey, Julia A. Mundy, David Goldhaber-Gordon, Jennifer E. Hoffman, Jason D. Hoffman, SmB_6 films shorted by B-doped Si substrates *(in preparation)*

12. Donovan L. Buterakos, **Sandesh S. Kalantre**, Joshua Ziegler, Jacob M. Taylor, Justyna P. Zwolak, QDFlow: A Python package for physics simulations of quantum dot devices, *SciPost Phys. Codebases*, 65 (2026)

11. Mihir Pendharkar, Steven J. Tran, Gregory Zaborski Jr., Joe Finney, Aaron L. Sharpe, Rupini V. Kamat, **Sandesh S. Kalantre**, et. al. and Marc A. Kastner, Andrew J. Mannix, David Goldhaber-Gordon, Torsional force microscopy of van der Waals moirés and atomic lattices, *Proceedings of the National Academy of Sciences* 121 (10), e2314083121, 2024

10. Joshua Ziegler, Thomas McJunkin, Emily S. Joseph, **Sandesh S. Kalantre**, Benjamin Harpt, Donald E. Savage, et. al. and Mark A. Eriksson, Jacob M. Taylor, Justyna P Zwolak, Toward robust autotuning of noisy quantum dot devices, *Physical Review Applied* 17 (2), 024069, 2022

9. Brian J. Weber, **Sandesh S. Kalantre**, Thomas McJunkin, Jacob M. Taylor, Justyna P. Zwolak, Theoretical bounds on data requirements for the ray-based classification, *SN Computer Science* 3 (1), 57, 2022

8. Christie J. Trimble, Ming-Tso Wei, Noah F. Q. Yuan, **Sandesh S. Kalantre** et. al. and James R. Williams, Josephson Detection of Time Reversal Symmetry Breaking Superconductivity in SnTe Nanowires, *Nature Partner Journal (npj), Quantum Materials* 6 (1), 1-6, 2021

7. **Sandesh S. Kalantre**, Fan Yu, Ming-Tso Wei, et. al. and James R. Williams, Anomalous Phase Dynamics of Driven Graphene Josephson junctions, *Physical Review Research* 2 (2), 023093, 2020

6. Justyna P. Zwolak, Thomas McJunkin, **Sandesh S. Kalantre**, J.P. Dodson, Evan R. MacQuarrie, et. al. and Mark A. Eriksson, Jacob M. Taylor, Autotuning of Double-Dot Devices In Situ with Machine Learning, *Physical Review Applied* 13 (3), 034075, 2020

5. Justyna P. Zwolak, Thomas McJunkin, **Sandesh S. Kalantre**, Samuel F. Neyens, Evan R. MacQuarrie, Mark A. Eriksson, Jacob M. Taylor, Ray-based framework for state identification in quantum dot devices, PRX Quantum 2 (2), 020335, 2021

4. Justyna P. Zwolak, **Sandesh S. Kalantre**, Thomas McJunkin, Brian J. Weber, Jacob M. Taylor, Ray-based classification framework for high-dimensional data, arXiv preprint arXiv:2010.00500, Proceedings of the Machine Learning and the Physical Sciences Workshop at *NeurIPS 2020*, Vancouver, Canada

3. **Sandesh S. Kalantre**, Justyna P. Zwolak, et. al. and Jacob M. Taylor, Machine learning techniques for state recognition and auto-tuning in quantum dots, *Nature Partner Journal (npj), Quantum Information 5 (1), 1-10, 2019*

2. Praveen Sriram, **Sandesh S. Kalantre**, Kaveh Gharavi, Jonathan Baugh, Bhaskaran Muralidharan, Supercurrent interference in semiconductor nanowire Josephson junctions, *Physical Review B* 100 (15), 155431, 2019

1. Justyna P. Zwolak, **Sandesh S. Kalantre**, Xingyao Wu, Stephen Ragole, Jacob M. Taylor, QFlow lite dataset: A machine-learning approach to the charge states in quantum dot experiments, *PloS one* 13 (10), e0205844, 2018

PATENTS

❖ Ray-based Classification Framework for Machine Learning-based Tuning Techniques *2023*
J.P. Zwolak, **S.S. Kalantre**, J.M. Taylor, T.W. McJunkin, US Patent App. 18/028,154

INVITED PRESENTATIONS

1. What can microwave resonators tell us about correlated materials?, AMO/QI Seminar, UC Berkeley, May 2026

2. Microwave resonance in thin films of spin liquid candidate $TbInO_3$, 7th International Workshop on Complex Oxides, Cargese, France, March 2024

CONTRIBUTED CONFERENCE PRESENTATIONS

1. Hybrid superconducting circuit architecture to probe van der Waals heterostructures, APS March Meeting, Denver, CO, March 2026

2. Microwave resonance probes of frustrated magnets, Capri School on Physics of Superconducting Devices, Capri, Italy, April 2025

3. Flip-chip spectroscopy of frustrated magnet $Gd_2Ti_2O_7$, MURI program on Interfacial Superconductivity, Annual Review, Stanford University, Dec 2024

4. Low temperature magneto-transport of epitaxial samarium hexaboride films, APS March Meeting, March 2024

5. Low temperature magnetotransport of SmB_6 films, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2023

6. Electron spin resonance measurements of spin liquid candidate TbInO3 with superconducting resonators, APS March Meeting, March 2023

7. Electron spin resonance measurements of spin liquid candidate TbInO3 with superconducting resonators, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2022

8. Electron Spin Resonance near the Metal Insulator Transition in Phosphorous doped Si, APS March Meeting, APS March Meeting, March 2022
9. Anomalous Phase Dynamics of graphene Josephson Junctions, APS March Meeting, APS March Meeting, March 2020
10. In-situ machine learning assisted auto-tuning of quantum dot devices, NISQ, UMD, June 2019
11. Machine Learning Techniques for quantum dots, JQI-QuICS-CMTC seminar, UMD, Dec 2018
12. Non-linear dynamics in graphene Josephson junctions, APS MAS, UMD, Nov 2018
13. ML and Quantum Dots, APS March Meeting, Los Angeles, CA, March 2018
14. ML and Quantum Dots, Quantum Seminar, NIST and UMD, July 2017

RESEARCH

Geballe Lab for Advanced Materials, Stanford University June 2022 - Present
Advisor: Prof. David Goldhaber-Gordon

- Engineered a NbTiN superconducting resonator platform with flip-chip sample integration for spin resonance characterization of novel frustrated magnets and quantum spin liquid candidates.
- Designed and validated a cQED microwave architecture for superfluid stiffness extraction in van der Waals superconductors, demonstrating GHz-frequency materials characterization at mK temperatures.
- Quantified topological transport signatures in SmB_6 thin films and $Eu_3In_2As_4$ nanowires, building a magneto-transport characterization pipeline for topological material candidates.

Quantum Materials Lab (QMDLab), University of Maryland August 2018 - June 2022
Advisor: Prof. James R. Williams

- Engineered superconducting interfaces in topological materials and van der Waals heterostructures through multi-modal electrical characterization, combining quasi-DC transport and microwave Shapiro response measurements.
- Built an analytical modeling framework for Josephson junction dynamics under microwave drive, leveraging non-linear oscillator theory to extract device parameters and predict Shapiro step signatures.

National Institute for Standards and Technology - Gaithersburg May 2017 - July 2017
Advisor: Prof. Jacob M. Taylor

- Developed CNN-based classification models for automated charge state recognition in semiconductor quantum dots, directly enabling downstream closed-loop device control.
- Engineered one of the first auto-tuning algorithms for quantum dots, autonomously driving an in-situ device to the single-electron regime — a first-of-its-kind demonstration in the field.

Indian Institute of Technology, Bombay August 2016 - May 2018
Advisor: Prof. Suddhasatta Mahapatra

- Nano-fabrication of nano-scale devices in semiconductor heterostructures (SiGe)
- Worked on a theoretical problem of Electron Dipole Spin Resonance (EDSR) selectivity of Phosphorous donor qubits in micromagnet fields

Institute for Science and Technology (IST) - Austria, Vienna May 2016 - July 2016
Advisor: Dr. Giorgos Katsaros

- Studied theoretical aspects of transport in quantum dot and nanowire systems in semi-conductors
- Worked on conductance measurements to observe Coulomb blockade and diamonds in p-type transistors at 4 K cooled with liquid He

Indian Institute of Technology, Bombay August 2016 - May 2018
Advisor: Prof. Bhaskaran Muralidharan

- Non-equilibrium Green's function (NEGF) formalism to model electron transport in superconductor-semiconductor systems
- Determined the effects of angular momentum subbands and Andreev Bound states to the current-phase relationship in nanowires connected to superconducting contacts

RESEARCH SKILLS

Quantum Device Fabrication

Magnetron sputtering	E-beam evaporation	Atomic layer deposition	Photolithography
E-beam lithography	Reactive ion etching	AFM imaging	SEM imaging
2D materials	Josephson junctions	2D Stacking	μ-wave resonators

Cryogenic & Microwave Instrumentation

Dilution refrigerators	VTI	VNA spectroscopy	Quasi-DC transport
RF reflectometry	Custom mixer circuits	High-speed digitizers	Lock-in amplifiers

Simulation & Computational Tools

Python	C++	Qcodes	gdspy/gdstk
NumPy	SciPy	QuTiP	TensorFlow
Sonnet	HFSS	COMSOL	SolidWorks
FreeCAD	Kicad	Metal machining	3D printing

TEACHING EXPERIENCE

Teaching Assistant, Stanford University *2023-2025*

- TA for undergraduate electricity and magnetism and quantum mechanics II
- Mentored student teams in the design of a nuclear quadrupole resonance setup & cryogenic probes for advanced undergraduate physics lab

Teaching Assistant, IIT Bombay *2015-2018*

- TA for a range of physics & mathematics undergraduate courses on electricity and magnetism, introductory quantum physics and partial-differential equations; on two occasions was also the head TA responsible for course logistics and personnel management

LEADERSHIP EXPERIENCE

Organizational Volunteer and Board Member, USAAAO *2018-Present*

- Designed astronomy and astrophysics problems at the high-school level, and assisted in coaching the US national team to the International Olympiad on Astronomy and Astrophysics (IOAA)

General Secretary, Department of Physics, IIT Bombay *2017-2018*

- Elected as the top student representative of the physics department's bachelor and masters students who communicated between students, faculty and other representatives at the institute level; work led to curriculum changes and bridged the gap in student-faculty communication in the department

Institute Student Mentor, IIT Bombay 2017-2018

- Senior mentor to a wing of 12 freshmen to assist them in settling to life at a university

Academic Resource Person, International Physics Olympiad (IPhO), Mumbai 2015

- Exam grader; worked in a team of 2 in development of IPho-rum — a browser based application for tasks such as voting, translation upload and feedback submission among $\approx$ *100 users*

MISCELLANEOUS INTERESTS

- Science popularization at IIT Bombay at the undergraduate-level, gave talks on varied topics such as Celestial mechanics and N-body problem, Cosmology and Quantum Computing
- Violin and classical music
- Won a Bronze medal in Astronomy hackathon as part of InterIIT Tech Meet
- Secured first place with a team of five in *Online Physics Brawl 2016*
- Teaching volunteer at NGOs for underprivileged children in Powai, Mumbai

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